Eine Woche, ein Beispiel

1.21. complex multilinear algebra

The title comes from
http://staff.ustc.edu.cn/~wangzuoq/Courses/16F-Manifolds/Notes/Lec16.pdf

We also take the reference from "Introduction to complex geometry", written by Yalong Shi:
http://maths.nju.edu.cn/~yshi/BICMR_ComplexGeometry.pdf

M: cplx mfld, $p \in M$ $\qquad$ $M_{\mathbb{R}}$: M viewed as smooth mfld, not base change

better: M_{sm}

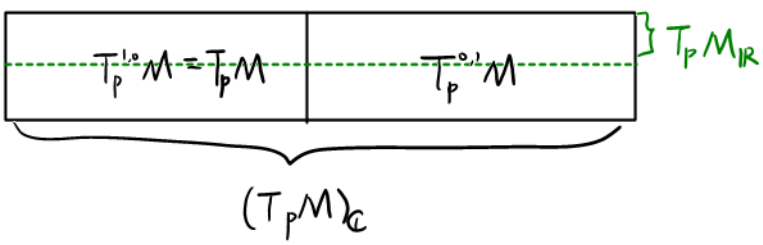
e.g. $M = \mathbb{C}^3$ $\quad p = 0$

Notation	base field	dim	basis		name	[YS20]
T_pM	$\mathbb{C}$	3	$\frac{\partial}{\partial z_i}$	holomorphic	tangent vector	
$T_pM_{\mathbb{R}}$	$\mathbb{R}$	6	$\frac{\partial}{\partial x_i}, \frac{\partial}{\partial y_i}$	real	tangent vector	$T_p^{\mathbb{R}}M$
$(T_pM)_{\mathbb{C}} := T_pM_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$	$\mathbb{C}$	6	$\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_i}$ or $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$	complexified	tangent vector	$T_p^{\mathbb{C}}M$
$T_p^{1,0}M = T_pM$	$\mathbb{C}$	3	$\frac{\partial}{\partial z_i}$	holomorphic	tangent vector	
$T_p^{0,1}M$	$\mathbb{C}$	3	$\frac{\partial}{\partial \bar{z}_i}$	anti-holomorphic	tangent vector	
T_p^*M	$\mathbb{C}$	3	dz_i	holomorphic	1-form	Ω_p^1
$T_p^*M_{\mathbb{R}} \triangleq \Omega^1_{\mathbb{R},p}$	$\mathbb{R}$	6	dx_i, dy_i	real	1-form	
$(T_p^*M)_{\mathbb{C}} := T_p^*M_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$	$\mathbb{C}$	6	$dz_i, d\bar{z}_i$ or dx, dy	complexified	1-form	$T_p^{*\mathbb{C}}M = A_p^1$
$T_p^{1,0,*}M \triangleq \Omega_p^{1,0} = T_p^*M$	$\mathbb{C}$	3	dz_i	(1,0)-form		$T_p^{*1,0}M = A_p^{1,0}$
$T_p^{0,1,*}M \triangleq \Omega_p^{1,0}$	$\mathbb{C}$	3	$d\bar{z}_i$	(0,1)-form		$T_p^{*0,1}M = A_p^{0,1}$

$\Omega^i, \Omega^{i,j}$: sheaves on M

Rmk. We don't have any natural identification between T_pM & $T_pM_{\mathbb{R}}$.
Notice that $\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$, $-\frac{1}{2}i$ is not real, so $\frac{\partial}{\partial z} \notin T_pM_{\mathbb{R}}$.

although our geometrical intuition of T_pM is often $T_pM_{\mathbb{R}}$,
$T_pM \cap T_pM_{\mathbb{R}} = \emptyset$ in $(T_pM)_{\mathbb{C}}$.



Reminder: the (induced) almost complex structure is defined as

$$J: T_pM_{\mathbb{R}} \longrightarrow T_pM_{\mathbb{R}}$$
$$\frac{\partial}{\partial x_i} \longmapsto \frac{\partial}{\partial y_i}$$
$$\frac{\partial}{\partial y_i} \longmapsto -\frac{\partial}{\partial x_i}$$

$$\rightsquigarrow \quad J: T_pM \longrightarrow T_pM$$

$$J\left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial y_i}\right) = \left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial y_i}\right)\begin{pmatrix} & -1 \\ 1 & \end{pmatrix}$$

$$J\left(\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_i}\right) = \left(\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_i}\right)\begin{pmatrix} i & \\ & -i \end{pmatrix}$$

real basis of $(T_pM)_{\mathbb{C}}$:

$$\mathcal{B}_1 = \{\underbrace{\partial_x, \partial_y}, \underbrace{i\partial_x, i\partial_y}\}$$

$$\circlearrowleft T_pM_{\mathbb{R}} \longleftrightarrow iT_pM_{\mathbb{R}} \circlearrowleft$$

$$\circlearrowleft_{Id} \qquad \circlearrowleft_{-Id}$$

$$\begin{cases} \partial_x = \partial_z + \partial_{\bar{z}} \\ \partial_y = \frac{1}{i}(\partial_z - \partial_{\bar{z}}) \end{cases} \qquad \begin{aligned} dx &= \tfrac{1}{2}(dz + d\bar{z}) \\ dy &= \tfrac{1}{2i}(dz - d\bar{z}) \end{aligned}$$

$$\mathcal{B}_2 = \{\underbrace{\partial_z}, \underbrace{\partial_{\bar{z}}}, \underbrace{i\partial_z}, \underbrace{i\partial_{\bar{z}}}\}$$

$$\circlearrowleft T_pM \longleftrightarrow T^{0,1}M \circlearrowleft$$

$$\circlearrowleft_{\times i} \qquad \circlearrowleft_{\times i}$$

$$\begin{cases} \partial_z = \frac{1}{2}(\partial_x - i\partial_y) \\ \partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y) \end{cases} \qquad \begin{aligned} dz &= dx + i\,dy \\ d\bar{z} &= dx - i\,dy \end{aligned}$$

$\circlearrowleft$: J, conj_J

$\circlearrowleft$: conj_i

$\circlearrowleft$: $\times i$

⚠ The conjugation usually dont preserve cplx subspace.

E.g. in $\mathbb{C}^2$, $\mathbb{C}\cdot(1, i)$ is not preserved by conjugation.

⚠ A cplx v.s don't necessary have a canonical conjugation structure, unless we identify a real structure.
In our case, conj_J is not canonically defined.
We take the real structure $\langle \partial_z, \partial_{\bar{z}} \rangle_{\mathbb{R}}$, which is not canonical.
change of coordinates will change this.

Rmk. $V = \mathbb{C}^n$

$$\{\text{conjugations of } V\} \longleftrightarrow \{\text{dec } V = W \oplus iW\} \cong GL_n(\mathbb{C})/GL_n(\mathbb{R})$$
$$\sigma \longmapsto V = V^\sigma \oplus iV^\sigma$$

$$(T_pM)_{\mathbb{C}}^{\mathrm{conj}_i} = T_pM_{\mathbb{R}}$$

$$(T_pM)_{\mathbb{C}}^{\mathrm{conj}_J} = \langle \partial_x, i\partial_y \rangle_{\mathbb{R}} = \langle \partial_z, \partial_{\bar{z}} \rangle_{\mathbb{R}}$$ not stable under i-action

$$\begin{pmatrix} 1 & \\ & i \end{pmatrix}\begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2}i \\ \frac{1}{2} & \frac{1}{2}i \end{pmatrix}$$ in $GL_n(\mathbb{C})/GL_n(\mathbb{R})$

When viewed as cplx v.s.
the natural conjugation on $T_p^{p,q}M$ is induced by conj_i.

$$
\begin{aligned}
J\,(f\partial_x + g\partial_y) &= f\partial_y - g\partial_x \\
J\,(\partial_x, \partial_y, i\partial_x, i\partial_y) &= (\partial_y, -\partial_x, i\partial_y, -i\partial_x) \\
&= (\partial_x, \partial_y, i\partial_x, i\partial_y) \begin{pmatrix} & -1 & & \\ 1 & & & \\ & & & -1 \\ & & 1 & \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
J\,(f\partial_z + g\partial_{\bar z}) &= if\partial_z - ig\partial_{\bar z} \\
J\,(\partial_z, \partial_{\bar z}, i\partial_z, i\partial_{\bar z}) &= (i\partial_z, -i\partial_{\bar z}, -\partial_z, \partial_{\bar z}) \\
&= (\partial_z, \partial_{\bar z}, i\partial_z, i\partial_{\bar z}) \begin{pmatrix} & & -1 & \\ & & & 1 \\ 1 & & & \\ & -1 & & \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{conj}_J\,(f\partial_x + g\partial_y) &= \bar f\partial_x - \bar g\partial_y \\
\mathrm{conj}_J\,(\partial_x, \partial_y, i\partial_x, i\partial_y) &= (\partial_x, -\partial_y, -i\partial_x, i\partial_y) \\
&= (\partial_x, \partial_y, i\partial_x, i\partial_y) \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & 1 \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{conj}_J\,(f\partial_z + g\partial_{\bar z}) &= \bar f\partial_z + \bar g\partial_{\bar z} \\
\mathrm{conj}_J\,(\partial_z, \partial_{\bar z}, i\partial_z, i\partial_{\bar z}) &= (\partial_z, \partial_{\bar z}, -i\partial_z, -i\partial_{\bar z}) \\
&= (\partial_z, \partial_{\bar z}, i\partial_z, i\partial_{\bar z}) \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{conj}_i\,(f\partial_x + g\partial_y) &= \bar f\partial_x + \bar g\partial_y \\
\mathrm{conj}_i\,(\partial_x, \partial_y, i\partial_x, i\partial_y) &= (\partial_x, \partial_y, -i\partial_x, -i\partial_y) \\
&= (\partial_x, \partial_y, i\partial_x, i\partial_y) \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{conj}_i\,(f\partial_z + g\partial_{\bar z}) &= \bar f\partial_{\bar z} + \bar g\partial_z \\
\mathrm{conj}_i\,(\partial_z, \partial_{\bar z}, i\partial_z, i\partial_{\bar z}) &= (\partial_{\bar z}, \partial_z, -i\partial_{\bar z}, -i\partial_z) \\
&= (\partial_z, \partial_{\bar z}, i\partial_z, i\partial_{\bar z}) \begin{pmatrix} & 1 & & \\ 1 & & & \\ & & & -1 \\ & & -1 & \end{pmatrix}
\end{aligned}
$$

Rmk. The J-action on $T_0^* X_{\mathbb{R}}$ comes from the following iso:

$$\begin{tikzcd} T_0^* X_{\mathbb{R}} \arrow[r, hook] \arrow[loop below, "J"] & T_0^* X_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \arrow[r, "\cong"] \arrow[loop below, "J"] \arrow[loop right, "i"] & T_0^* X \oplus T_0^{0,1} X \arrow[r, two heads] & T_0^* X \arrow[loop right, "i"] \end{tikzcd}$$

Take the albanese variety, we have J-action on $H^1(X;\mathbb{R})$:

$$\begin{tikzcd} H^1(X;\mathbb{R}) \arrow[r, hook] \arrow[loop below, "J"] & H^1(X;\mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C} \arrow[r, "\cong"] \arrow[loop below, "J"] \arrow[loop right, "i"] & H^{1,0} \oplus H^{0,1} \arrow[r, two heads] & H^{1,0} = H^0(X;\Omega_X^1) \arrow[loop right, "i"] \end{tikzcd}$$

Q: Which cohomology can J act naturally? $H^n_{\text{betti}}(X;\mathbb{C})$? $H^n_{dR}(X;\mathbb{R})$? $H^n_{dR}(X/\mathbb{R})$?

When is J-action on cohomology defined via the J-action on differential forms?

My guess:

$$J \text{ acts on } \begin{cases} H^1_{dR}(X;\mathbb{R}) & \text{via Albanese} \\ H^n_{dR}(X;\mathbb{C}) & \text{via Hodge dec} \end{cases}$$

or J preserves the harmonic form

However, since $J \circ d \neq d \circ J$, J does not act on $H^n_{dR}(X;\mathbb{R})$.

reality check:

$$\begin{tikzcd} \Omega^k \arrow[r, "d"] \arrow[d, "J"'] & \Omega^{k+1} \arrow[d, "J"] \\ \Omega^k \arrow[r, "d"] & \Omega^{k+1} \end{tikzcd} \qquad \begin{tikzcd} f\, dx_I dy_{I'} \arrow[r, mapsto] \arrow[d, mapsto] & \left(\sum_a \frac{\partial f}{\partial x_a} dx_a + \sum_b \frac{\partial f}{\partial y_b} dy_b\right) dx_I dy_{I'} \arrow[d, mapsto] \\ f\, dy_I dx_{I'} \arrow[r, mapsto] & \left(\sum_a \frac{\partial f}{\partial x_a} dy_a + \sum_b \frac{\partial f}{\partial y_b} dx_b\right) dy_I dx_{I'} \\ & \left(\sum_a \frac{\partial f}{\partial x_a} dx_a + \sum_b \frac{\partial f}{\partial y_b} dy_b\right) dy_I dx_{I'} \end{tikzcd}$$

Rmk

In fact, it is a bit subtle. In
[0812.3680v2, Thm 2.3, Prop 2.8]: Symplectic forms and cohomology decomposition of almost complex 4-manifolds, https://arxiv.org/abs/0812.3680v2

They define

(X, J): almost cplx $2n$-fold

$$H_J^+(X) := \{ [\alpha] \in H^n(X;\mathbb{R}) \mid J\alpha = \alpha \}$$
$$H_J^-(X) := \{ [\alpha] \in H^n(X;\mathbb{R}) \mid J\alpha = -\alpha \}$$

and show that

- When X is cpt & $n=2$,
$$H^2(X;\mathbb{R}) = H_J^+(X) \oplus H_J^-(X);$$
- When X is cpt & J admits a compatible sympletic structure,
$$H^n(X;\mathbb{R}) \supseteq H_J^+(X) \oplus H_J^-(X).$$ C^∞-pure

Hermitian metric

$$
\begin{aligned}
\overset{=ds^2}{H} &= h_{\alpha\beta}\, dz^\alpha \otimes d\bar{z}^\beta && \text{Hermitian metric} \quad (h_{\alpha\beta}) \in \mathbb{R}^{n\times n} \text{ pos def} \\
g &= \tfrac{1}{2}(H + \overline{H}) && \text{Riemannian metric} \\
\omega &= \tfrac{i}{2}(H - \overline{H}) && \text{Hermitian form}
\end{aligned}
$$

e.g.

$$
\begin{aligned}
H &= dz \otimes d\bar{z} \\
&= (dx \otimes dx + dy \otimes dy) - i(dx \otimes dy - dy \otimes dx) = g - i\omega \\
g &= \tfrac{1}{2}(dz \otimes d\bar{z} + d\bar{z} \otimes dz) \\
&= dx \otimes dx + dy \otimes dy \\
\omega &= \tfrac{i}{2}(dz \otimes d\bar{z} - d\bar{z} \otimes dz) && = i\, dz \wedge d\bar{z} \\
&= dx \otimes dy - dy \otimes dx && = 2\, dx \wedge dy
\end{aligned}
$$

$$
\kappa = -\frac{1}{h}\partial_z \partial_{\bar{z}} \ln h = -\frac{1}{h}\frac{1}{4}\Delta(\ln h) \qquad \Delta = \partial_x^2 + \partial_y^2
$$

Two methods to show $\mathbb{I}(\partial_z, \partial_{\bar{z}}) = 0$

Method 1.

$$
i\,\mathbb{I}(\partial_z, \partial_{\bar{z}}) = \mathbb{I}(J\partial_z, \partial_{\bar{z}}) = \mathbb{I}(\partial_z, J\partial_{\bar{z}}) = -i\,\mathbb{I}(\partial_z, \partial_{\bar{z}})
$$

Method 2.

$$
\begin{aligned}
\mathbb{I}(\partial_z, \partial_{\bar{z}}) &= \mathbb{I}\left(\tfrac{1}{2}(\partial_x - i\partial_y), \tfrac{1}{2}(\partial_x + i\partial_y)\right) \\
&= \mathbb{I}\left(\tfrac{1}{2}(1 - iJ)\partial_x, \tfrac{1}{2}(1 + iJ)\partial_x\right) \\
&= \tfrac{1}{2}(1 - iJ) \cdot \tfrac{1}{2}(1 + iJ)\, \mathbb{I}(\partial_x, \partial_x) \\
&= \tfrac{1}{4}(1 - i^2 J^2)\, \mathbb{I}(\partial_x, \partial_x) \\
&= 0.
\end{aligned}
$$